

He (Harvey) Li

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Atlanta, GA

Education

Georgia Institute of Technology | GPA: 3.92
B.S. | Computer Science - Intelligence & Cybersecurity

Jul 2025 - Expected May 2029
Atlanta, GA

Publications

Chung-En Ho, Weiyu Sun, Cheng-Jhih Shih, **He Li**, Yong Liu, Yingyan Celine Lin, “SpecFold: Folding Cross-Branch Redundancy for Faster Speculative Decoding in Diffusion Language Models”, *ICLR*, 2027. **In progress**
Wenhao Chen, **He Li**, Shen Gang, Rundong Cao, Guojie Song, “DeltaAug: A Directional Semantic Data Augmentation Framework Guided by Differential Reasoning”, *PAKDD*, 2025.

Experience

Student Researcher

Sep 2025 - May 2026

EIC Lab, Georgia Institute of Technology

Atlanta, GA

- Identified latent cross-branch redundancy in the speculative decoding trees of diffusion LLMs and characterized it through numerical experiments across LLaDA and Fast-dLLM.
- Designed the reuse policy at the core of SpecFold, which cuts up to 62% of FLOPs over Spiffy while largely preserving task performance; drafted its initial CUDA implementation and wrote the paper.
- Programmed custom CUDA kernels for speculative decoding and K-V cache management in dLLMs, plus hardware-software co-designs targeting custom FPGA accelerators.

ML Researcher

May 2026 - present

Respan.ai (YC W24)

San Francisco, CA

- Proposed and designed [Span-01](#), a classifier that detects production LLM agent behaviors (jailbreaks, hallucinations, failures) in a single forward pass via a hybrid linear-attention architecture with parallel branches, trained with RLAIIF to beat GPT-6, Luna, and Jev on F1; built its Rust inference router (harvey-l.com/blog/span-1-inference-router) with Clockwork-style SLO-aware scheduling and CUDA/Metal kernels across heterogeneous GPU and Mac Mini nodes, reaching 0.88M tok/sec and a 6.3x p99 latency cut.
- Designed Respan P0, a PII detection model pairing a decision tree with a custom-trained NER model; best in class on AI4Privacy F1 at 79.5%, vs. AWS 69.1, Raindrop 44.9, Presidio 34.5.
- Built and open-sourced Respan Security (github.com/respanai/respan-redteam), a red-teaming framework implementing TAP/Crescendo attacks; remediated a P0 RCE in the workflow engine and led the post-mortem.
- Migrated the backend to K8s (tuned KEDA/Karpenter, spot-safe pods) and rewrote consumers, gateway, and ingestion in Rust, cutting the monthly bill from \$103k to \$30k.

Student Researcher

Jan 2023 - Jul 2025

Institute for Artificial Intelligence, Peking University

Beijing, CN

- Implemented the augmentation pipeline and ran the benchmark evaluations for DeltaAug (PAKDD 2025), a directional semantic data augmentation framework guided by differential reasoning; also led research on extreme quantization for low-bit LLMs and Diffusion Transformers.

Projects

Tacklbox: Data powered app for anglers

May 2026

[tackl.co](#)

- Co-founded Tackl Co (backed by Georgia Tech CREATE-X) and shipped Tacklbox end-to-end (React Native app, Python/FastAPI backend, feature set and UI), now at 3,000+ installs and 20K social media followers.
- Generated habitat-suitability data from HydroLakes and the General Lake Model, then trained a logistic-regression predictor on it to score locations in-app.
- Built a GPU-accelerated fish ID pipeline (YOLOv10 detection, FPN/ResNet18 segmentation, ConvNeXt-Tiny classification over 639 species) with 6x throughput via batching.

Gordian: The fastest AI PCB designer

2026 - present

[getgordian.com](#)

- Designed and built the whole system solo: a simulated-annealing schematic layout engine, a quadtree capacity-mesh PCB autorouter, and the agent harness driving them.
- Produces a DRC-validated STM32 Blue Pill in 52s; comparable tools (Trace, Flux.ai) fail after 10 minutes.

Skills

Expertise: Deep Learning & NLP, LLM Agents & Red Teaming, Efficient Inference, Concurrent Programming, Reverse Engineering

Languages: Rust, Go, Python, C/C++, TypeScript, Assembly (x86, ARM), Java/Kotlin, Bash

Software & Tools: PyTorch, CUDA, OpenCV, React/NextJS, React Native, Docker and K8s, Linux, Git, IDA Pro, Frida, QEMU